

MARINE HVAC SYSTEM DESIGN CALCULATION REPORT

MV ANAMIKA — 45 ft Kerala Houseboat

Alappuzha (Alleppey) Backwaters, Kerala, India

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Calculation Standard: ASHRAE Fundamentals / CIBSE Guide

Refrigerant Design: ASHRAE 15 / EN 378

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1. VESSEL SPECIFICATION & DESIGN PARAMETERS

1.1 Vessel Description

MV Anamika is a 45-foot luxury houseboat designed for charter operations on the Kerala backwaters. The vessel accommodates 6 guests and 2 crew, with continuous operation under tropical conditions.

Parameter	Value	Remarks
Length Overall (LOA)	13.7 m (45 ft)	Hull length
Beam (max)	4.3 m (14.1 ft)	External width
Interior headroom	2.1 m	Saloon and cabins
Freeboard (deck to WL)	0.3 m	Loaded condition
Draft	0.6 m	Loaded waterline
Total hull height	2.4 m	Freeboard + draft
Max occupancy	8 persons	6 guests + 2 crew
Total conditioned area	56.0 m ²	Sum of all zones
Conditioned volume	117.6 m ³	56.0 × 2.1 m
Hull material	8 mm GRP (fiberglass)	Glass reinforced plastic
Deck material	20 mm teak / 8 mm GRP	Teak laid over GRP
Topsides insulation	25 mm polyurethane foam	Inside GRP topsides
Deck insulation	50 mm polyurethane foam	Between GRP and ply
Operational location	Alappuzha, Kerala (9.5 °N)	Vembanad Lake area
Power supply	15 kVA diesel generator	≈ 12 kW usable at 0.80 pf

1.2 Zone Layout

Zone	L (m)	W (m)	Floor Area (m ²)	Occupancy	Use
Main Saloon	4.5	4.3	19.4	5 persons	Dining / lounge
Master Cabin	3.5	4.3	15.1	2 persons	Sleeping
Cabin 2	3.0	4.3	12.9	2 persons	Sleeping
Galley	2.0	4.3	8.6	1 crew	Cooking / prep
Total	—	—	56.0	8 persons	—

1.3 Design Conditions

Parameter	Outdoor	Indoor (Set-Point)
Dry Bulb Temperature	35 °C (ASHRAE 0.4% cooling)	24 °C
Wet Bulb Temperature	28 °C	—
Relative Humidity	75 %	50 %

Humidity Ratio (ω)	0.02703 kg/kg	0.00930 kg/kg
Backwater temperature	30 °C (measured)	—
Peak solar (horiz.)	700 W/m ² (Kerala, April)	—
Peak solar (vertical)	300 W/m ² (avg. all sides)	—
Location latitude	9.5 °N	—

Humidity ratio verification (from psychrometric equations):

At 35 °C, 75 % RH:

$$P_{\text{sat}}(35^{\circ}\text{C}) = 5.628 \text{ kPa} \quad [\text{Antoine equation}]$$

$$P_w = 0.75 \times 5.628 = 4.221 \text{ kPa}$$

$$\omega_{\text{out}} = 0.622 \times P_w / (P_{\text{atm}} - P_w) = 0.622 \times 4.221 / (101.325 - 4.221)$$

$$\omega_{\text{out}} = 0.622 \times 4.221 / 97.104 = 0.02703 \text{ kg/kg}$$

At 24 °C, 50 % RH:

$$P_{\text{sat}}(24^{\circ}\text{C}) = 2.985 \text{ kPa}$$

$$P_w = 0.50 \times 2.985 = 1.4925 \text{ kPa}$$

$$\omega_{\text{in}} = 0.622 \times 1.4925 / (101.325 - 1.4925) = 0.00930 \text{ kg/kg}$$

$$\Delta\omega = 0.02703 - 0.00930 = 0.01773 \text{ kg/kg} \quad | \quad \Delta T = 35 - 24 = 11^{\circ}\text{C}$$

2. THERMAL TRANSMITTANCE (U-VALUE) CALCULATIONS

The overall U-value for each construction element is determined using the standard thermal resistance summation method (ISO 6946). Each layer's resistance is: $R = \text{thickness} / \text{thermal conductivity} (t/k)$. Surface resistances are taken from ASHRAE Fundamentals Table.

2.1 Below Waterline Hull — 8 mm GRP

The hull below the waterline is submerged in backwater at 30 °C. The external convective resistance is very low due to water's high convective coefficient ($h \approx 500 \text{ W/m}^2\text{K}$).

$$\begin{aligned} R_{\text{total}} &= R_{\text{water-surface}} + R_{\text{GRP}} + R_{\text{int-surface}} \\ R_{\text{water-surface}} &= 1/h_{\text{water}} = 1/500 &= 0.0020 \text{ m}^2\text{K/W} \\ R_{\text{GRP}} &= t/k = 0.008/0.30 &= 0.0267 \text{ m}^2\text{K/W} \\ R_{\text{int-surface}} &(\text{still air, ASHRAE}) &= 0.1300 \text{ m}^2\text{K/W} \\ \hline R_{\text{total}} \text{ (BW)} &&= 0.1587 \text{ m}^2\text{K/W} \end{aligned}$$

$$| \quad U_{\text{BW}} = 1 / R_{\text{total}} = 1 / 0.1587 = 6.30 \text{ W/m}^2\text{K}$$

2.2 Above Waterline Hull — GRP + 25 mm PU Foam + 4 mm Ply Liner

The topsides are insulated from inside with closed-cell polyurethane foam ($k = 0.025 \text{ W/mK}$) finished with 4 mm marine plywood.

$$\begin{aligned} R_{\text{total}} &= R_{\text{ext}} + R_{\text{GRP}} + R_{\text{foam}} + R_{\text{ply}} + R_{\text{int}} \\ R_{\text{ext}} &(\text{forced air, ASHRAE}) &= 0.0400 \text{ m}^2\text{K/W} \\ R_{\text{GRP}} &= 0.008/0.30 &= 0.0267 \text{ m}^2\text{K/W} \\ R_{\text{foam}} &= 0.025/0.025 &= 1.0000 \text{ m}^2\text{K/W} \\ R_{\text{ply}} &= 0.004/0.13 &= 0.0308 \text{ m}^2\text{K/W} \\ R_{\text{int}} &&= 0.1300 \text{ m}^2\text{K/W} \\ \hline R_{\text{total}} \text{ (AW)} &&= 1.2275 \text{ m}^2\text{K/W} \end{aligned}$$

$$| \quad U_{\text{AW}} = 1 / 1.2275 = 0.815 \text{ W/m}^2\text{K}$$

2.3 Deck — 20 mm Teak + 8 mm GRP + 50 mm PU Foam + 4 mm Ply

The deck receives direct solar radiation. The insulation layer is 50 mm (double the topsides) to account for the high solar load on the horizontal surface.

$$\begin{aligned} R_{\text{total}} &= R_{\text{ext}} + R_{\text{teak}} + R_{\text{GRP}} + R_{\text{foam}} + R_{\text{ply}} + R_{\text{int}} \\ R_{\text{ext}} &&= 0.0400 \text{ m}^2\text{K/W} \\ R_{\text{teak}} &= 0.020/0.17 &= 0.1176 \text{ m}^2\text{K/W} \\ R_{\text{GRP}} &= 0.008/0.30 &= 0.0267 \text{ m}^2\text{K/W} \\ R_{\text{foam}} &= 0.050/0.025 &= 2.0000 \text{ m}^2\text{K/W} \\ R_{\text{ply}} &= 0.004/0.13 &= 0.0308 \text{ m}^2\text{K/W} \\ R_{\text{int}} &&= 0.1300 \text{ m}^2\text{K/W} \\ \hline R_{\text{total}} \text{ (Deck)} &&= 2.3451 \text{ m}^2\text{K/W} \end{aligned}$$

$$| \quad U_{\text{deck}} = 1 / 2.3451 = 0.426 \text{ W/m}^2\text{K}$$

2.4 Bulkheads — 12 mm Marine Ply + 50 mm Rockwool + 12 mm Ply

Interior partition bulkheads. Both surfaces have interior still-air resistance.

$$\begin{aligned} R_{\text{total}} &= 2 \times R_{\text{int}} + R_{\text{ply1}} + R_{\text{rockwool}} + R_{\text{ply2}} \\ 2 \times R_{\text{int}} &&= 2 \times 0.1300 = 0.2600 \text{ m}^2\text{K/W} \\ R_{\text{ply}} &= 0.012/0.13 &= 0.0923 \text{ m}^2\text{K/W} \text{ (each side)} \\ R_{\text{rockwool}} &= 0.050/0.035 &= 1.4286 \text{ m}^2\text{K/W} \\ \hline \end{aligned}$$

$$R_{\text{total}} \text{ (BH)} = 0.2600 + 0.0923 + 1.4286 + 0.0923 = 1.8732 \text{ m}^2\text{K/W}$$

$U_{\text{BH}} = 1 / 1.8732 = 0.534 \text{ W/m}^2\text{K}$

2.5 Glazing

Main saloon windows are marine-grade double-pane toughened glass. Portholes are single-pane acrylic. U-values taken from manufacturer data sheets (ASHRAE verified).

Glazing Type	Construction	U-Value (W/m²K)	SHGC
Main saloon windows	6 mm glass / 10 mm gap / 6 mm glass	2.80	0.40
Cabin portholes	10 mm acrylic, single-pane	5.50	0.70

2.6 Summary of U-Values

Construction Element	Build-Up	Total R (m²K/W)	U (W/m²K)
Below waterline hull	8mm GRP + water surface	0.1587	6.30
Above waterline hull	GRP + 25mm foam + 4mm ply	1.2275	0.815
Deck	20mm teak + GRP + 50mm foam + ply	2.3451	0.426
Bulkheads	12mm ply + 50mm rockwool + 12mm ply	1.8732	0.534
Main saloon windows	Double-pane marine glass	—	2.80
Cabin portholes	Single-pane acrylic	—	5.50

3. SURFACE AREA CALCULATIONS

All surface areas are derived from the vessel geometry defined in Section 1.

3.1 Hull Perimeter

$$\begin{aligned} \text{Perimeter (P)} &= 2 \times (\text{LOA} + \text{Beam}) = 2 \times (13.7 + 4.3) \\ P &= 2 \times 18.0 = 36.0 \text{ m} \end{aligned}$$

3.2 Deck Area

$$\begin{aligned} A_{\text{deck}} &= \text{LOA} \times \text{Beam} = 13.7 \times 4.3 \\ A_{\text{deck}} &= 58.91 \text{ m}^2 \end{aligned}$$

3.3 Above Waterline Hull Area

$$\begin{aligned} \text{Height}_{\text{AW}} &= \text{Total hull height} - \text{draft} = 2.4 - 0.6 = 1.8 \text{ m} \\ A_{\text{AW_gross}} &= P \times \text{Height}_{\text{AW}} = 36.0 \times 1.8 \\ A_{\text{AW_gross}} &= 64.80 \text{ m}^2 \end{aligned}$$

3.4 Below Waterline Hull Area

$$\begin{aligned} A_{\text{BW}} &= P \times \text{draft} = 36.0 \times 0.6 \\ A_{\text{BW}} &= 21.60 \text{ m}^2 \end{aligned}$$

3.5 Glazed Areas

Element	Count	Dimensions (m)	Area each (m ²)	Total Area (m ²)
Main saloon windows	3	0.8 × 0.6	0.48	1.44
Saloon portholes	4	0.2 × 0.3	0.06	0.24
Cabin portholes	4	0.2 × 0.3	0.06	0.24
Total glazing	—	—	—	1.92

3.6 Net Surface Areas (for heat gain calculations)

$$A_{\text{AW_net}} = A_{\text{AW_gross}} - A_{\text{glazing}} = 64.80 - 1.92 = 62.88 \text{ m}^2$$

| Deck: 58.91 m² | AW Hull (net): 62.88 m² | BW Hull: 21.60 m² | Glazing: 1.92 m²

4. SOL-AIR TEMPERATURE CALCULATION

The sol-air temperature (T_{sa}) is a fictitious outdoor temperature that accounts for both convective heat exchange with the ambient air and radiative heat gain from solar irradiance, combined into a single equivalent driving temperature for conduction calculations (ASHRAE Fundamentals Chapter 18).

$$T_{sa} = T_{out} + (\alpha \times I_{sol}) / h_{out} - (\varepsilon \times \Delta R_{LW}) / h_{out}$$

Where:

α = solar absorptivity of surface
 I_{sol} = total solar irradiance on surface (W/m^2)
 h_{out} = external surface heat transfer coefficient = $17 W/m^2K$ (ASHRAE, combined)
 ε = long-wave emissivity of surface
 ΔR_{LW} = long-wave radiation correction factor (W/m^2)
= $63 W/m^2$ for horizontal surface (ASHRAE)
= $0 W/m^2$ for vertical surface (ASHRAE, avg all orientations)

4.1 Deck (Horizontal Surface — Dark Teak)

α_{teak} = 0.90 (dark natural wood, measured absorptivity)
 I_{sol} = $700 W/m^2$ (Kerala April, peak horizontal irradiance, $9.5^\circ N$)
 h_{out} = $17 W/m^2K$
 ε = 0.90
 ΔR_{LW} = $63 W/m^2$ (horizontal, clear sky)
 T_{out} = $35^\circ C$

$$\begin{aligned} T_{sa}(\text{deck}) &= 35 + (0.90 \times 700)/17 - (0.90 \times 63)/17 \\ &= 35 + 630/17 - 56.7/17 \\ &= 35 + 37.06 - 3.33 \end{aligned}$$

| $T_{sa}(\text{deck}) = 68.73^\circ C$

4.2 Above Waterline Hull (Vertical, Average All Orientations)

For a vessel that can rotate at anchor, the effective average solar load on vertical surfaces is calculated using an all-orientation average. The long-wave correction for vertical surfaces averages to zero (ASHRAE).

α_{GRP} = 0.70 (cream/white painted GRP exterior)
 I_{v_avg} = $300 W/m^2$ (average across all orientations, Kerala summer)
 ΔR_{LW} = $0 W/m^2$ (vertical surface)
 $T_{sa}(\text{AW hull}) = 35 + (0.70 \times 300)/17 - 0$
= $35 + 210/17$
= $35 + 12.35$

| $T_{sa}(\text{AW hull}) = 47.35^\circ C$

5. COOLING LOAD CALCULATIONS

The cooling load is the rate of heat that must be extracted from the conditioned space to maintain the desired indoor conditions. Components include: transmission through the envelope, solar radiation through glazing, internal gains (occupancy, lighting, equipment), and the ventilation/infiltration load.

5.1 Transmission Heat Gains Through Envelope

5.1.1 Deck

Driven by the sol-air temperature calculated in Section 4.1.

$$\begin{aligned} Q_{\text{deck}} &= U_{\text{deck}} \times A_{\text{deck}} \times (T_{\text{sa}} - T_{\text{in}}) \\ &= 0.426 \times 58.91 \times (68.73 - 24) \\ &= 0.426 \times 58.91 \times 44.73 \end{aligned}$$

$$| \quad Q_{\text{deck}} = 1,122 \text{ W}$$

5.1.2 Above Waterline Hull (Net of Glazing)

$$\begin{aligned} Q_{\text{AW}} &= U_{\text{AW}} \times A_{\text{AW_net}} \times (T_{\text{sa,AW}} - T_{\text{in}}) \\ &= 0.815 \times 62.88 \times (47.35 - 24) \\ &= 0.815 \times 62.88 \times 23.35 \end{aligned}$$

$$| \quad Q_{\text{AW}} = 1,196 \text{ W}$$

5.1.3 Below Waterline Hull

The backwater temperature (30 °C) exceeds the indoor set-point (24 °C), so the below-waterline hull is a net heat GAIN, not a heat sink.

$$\begin{aligned} Q_{\text{BW}} &= U_{\text{BW}} \times A_{\text{BW}} \times (T_{\text{water}} - T_{\text{in}}) \\ &= 6.30 \times 21.60 \times (30 - 24) \\ &= 6.30 \times 21.60 \times 6 \end{aligned}$$

$$| \quad Q_{\text{BW}} = 816 \text{ W (heat gain from warm backwater)}$$

5.1.4 Windows — Conduction Component

The CLTD method (Cooling Load Temperature Difference) applies a correction for the diurnal temperature swing. CLTD for glass = 7 °C (ASHRAE, tropical correction applied).

$$\begin{aligned} Q_{\text{win_cond}} &= \Sigma U \times A \times \text{CLTD}_{\text{glass}} \\ \text{Saloon windows: } Q &= 2.80 \times 1.44 \times 7 = 28.2 \text{ W} \\ \text{Portholes: } Q &= 5.50 \times 0.48 \times 7 = 18.5 \text{ W} \end{aligned}$$

$$| \quad Q_{\text{win (conduction)}} = 46.7 \text{ W} \approx 47 \text{ W}$$

5.1.5 Windows — Solar Radiation (SHGC/CLF Method)

Solar heat gain through glass is calculated using: Solar Heat Gain Factor (SHGF), the glass Solar Heat Gain Coefficient (SHGC), and the Cooling Load Factor (CLF) which accounts for thermal mass in the space.

$$\begin{aligned} Q_{\text{solar}} &= \text{SHGC} \times A \times \text{SHGF}_{\text{max}} \times \text{CLF} \\ \text{Main saloon (SHGC=0.40, SHGF}_{\text{max}} &= 482 \text{ W/m}^2 \text{ for E-facing at } 9.5^\circ\text{N, CLF=0.65):} \\ Q &= 0.40 \times 1.44 \times 482 \times 0.65 = 180 \text{ W} \\ \text{Portholes (SHGC=0.70, SHGF}_{\text{avg}} &= 300 \text{ W/m}^2, \text{ CLF=0.65):} \\ Q &= 0.70 \times 0.48 \times 300 \times 0.65 = 65 \text{ W} \end{aligned}$$

$$| \quad Q_{\text{solar (glass)}} = 245 \text{ W}$$

5.2 Internal Heat Gains

5.2.1 Occupancy

Based on ASHRAE Fundamentals Table 1, Chapter 18 — light activity, typical comfort seating.

$$Q_{occ} = N_{persons} \times q_{sens} \quad (\text{sensible})$$

$$Q_{occ} = N_{persons} \times q_{lat} \quad (\text{latent})$$

$$\text{Sensible per person} = 75 \text{ W} \quad (24^{\circ}\text{C, light activity})$$

$$\text{Latent per person} = 55 \text{ W}$$

$$\text{Number of persons} = 8 \text{ (design maximum)}$$

$$Q_{occ_sensible} = 8 \times 75 = 600 \text{ W}$$

$$Q_{occ_latent} = 8 \times 55 = 440 \text{ W}$$

Q_occupancy: Sensible = 600 W | Latent = 440 W

5.2.2 Lighting

LED lighting throughout. Lighting power density (LPD) for marine hospitality spaces from ASHRAE 90.1 reference: 10 W/m².

$$Q_{lighting} = LPD \times A_{conditioned}$$
$$= 10 \times 56.0$$

Q_lighting = 560 W (sensible, 100%)

5.2.3 Equipment

Equipment	Zone	Power (W)
Television + AV system	Saloon	200
Marine refrigerator (12V, 120W rated)	Galley	100
Small appliances (kettle, toaster etc.)	Galley	150
Laptop / phone charging misc.	All	100
Total equipment	—	550

Q_equipment = 550 W (sensible)

5.2.4 Galley Cooking

A galley exhaust fan (800 m³/h hood) removes a portion of cooking heat. The fraction released to the conditioned space is estimated at 30-40% of total cooking load.

$$\text{Total cooking load (gas hob + oven equivalent)} = 2,000 \text{ W}$$

$$\text{Fraction released to space (with exhaust hood)} = 15\%$$

$$Q_{cook_sensible} = 0.15 \times 2,000 = 300 \text{ W}$$

$$Q_{cook_latent} = 0.075 \times 2,000 = 150 \text{ W}$$

Q_cooking: Sensible = 300 W | Latent = 150 W

5.3 Ventilation and Infiltration Loads

5.3.1 Fresh Air Ventilation

Minimum outdoor air supply is determined by ASHRAE Standard 62.1 — Ventilation for Acceptable Indoor Air Quality. For marine hospitality spaces, the requirement is 10 L/s per person.

$$Q_v = N \times q_{person} = 8 \times 10 = 80 \text{ L/s} = 0.080 \text{ m}^3/\text{s}$$

Sensible ventilation load:

$$Q_{vent_s} = \rho_{air} \times c_p \times Q_v \times \Delta T$$
$$= 1.20 \times 1006 \times 0.080 \times (35 - 24)$$
$$= 1.20 \times 1006 \times 0.080 \times 11$$
$$= 1,067 \text{ W}$$

Latent ventilation load:

$$\begin{aligned}
 Q_{\text{vent}_l} &= \rho_{\text{air}} \times h_{\text{fg}} \times Q_v \times \Delta\omega \\
 &= 1.20 \times 2,450,000 \times 0.080 \times 0.01773 \\
 &= 1.20 \times 2,450,000 \times 0.001418 \\
 &= 4,165 \text{ W}
 \end{aligned}$$

Q_ventilation: Sensible = 1,067 W | Latent = 4,165 W

5.3.2 Infiltration

Infiltration is estimated at 0.5 air changes per hour (ACH), appropriate for a well-sealed GRP vessel with sliding doors and marine-grade hatches.

$$Q_{\text{inf_flow}} = \text{ACH} \times V_{\text{space}} / 3600 = 0.5 \times 117.6 / 3600 = 0.01633 \text{ m}^3/\text{s}$$

$$Q_{\text{inf}_s} = \rho \times c_p \times Q_{\text{inf}} \times \Delta T = 1.20 \times 1006 \times 0.01633 \times 11 = 218 \text{ W}$$

$$Q_{\text{inf}_l} = \rho \times h_{\text{fg}} \times Q_{\text{inf}} \times \Delta\omega = 1.20 \times 2,450,000 \times 0.01633 \times 0.01773 = 853 \text{ W}$$

Q_infiltration: Sensible = 218 W | Latent = 853 W

5.4 Cooling Load Summary

Heat Gain Component	Sensible (W)	Latent (W)	Total (W)
Deck — solar-air	1,122	—	1,122
Above WL hull — solar-air	1,196	—	1,196
Below WL hull — warm water	816	—	816
Windows — conduction (CLTD)	47	—	47
Windows — solar radiation (SHGC)	245	—	245
Occupancy (8 persons)	600	440	1,040
Lighting (LED, 10 W/m²)	560	—	560
Equipment (AV, galley)	550	—	550
Galley cooking (with exhaust)	300	150	450
Ventilation — fresh air (80 L/s)	1,067	4,165	5,232
Infiltration (0.5 ACH)	218	853	1,071
Subtotal	6,721	5,608	12,329
Safety factor (10%)	672	561	1,233
Design Cooling Load	7,393	6,169	13,562

Total Cooling Load = 13,562 W ≈ 13.6 kW = 3.86 TR → Select 4.0 TR system

Sensible Heat Ratio (SHR) = 7,393 / 13,562 = 0.545

The SHR of 0.545 is typical for Kerala's extremely humid tropical climate. The low SHR confirms the need for deep cooling coils capable of significant moisture removal (dehumidification), and is a critical input to coil selection.

5.5 Heating Load Assessment

In Kerala's tropical climate, mechanical heating is rarely required. A brief calculation is performed to confirm whether a heating mode is necessary.

$$\text{Minimum outdoor temp (winter night, Alappuzha)} = 22 \text{ }^{\circ}\text{C}$$

$$\text{Indoor set-point} = 24 \text{ }^{\circ}\text{C} \rightarrow \Delta T_{\text{heat}} = 24 - 22 = 2 \text{ }^{\circ}\text{C}$$

$$Q_{\text{heat}} = [\sum U_i \times A_i] \times \Delta T + Q_{\text{vent_heat}}$$

Envelope: $[0.426 \times 58.91 + 0.815 \times 62.88 + 6.30 \times 21.60 + 2.80 \times 1.44 + 5.50 \times 0.48] \times 2$
= $[25.1 + 51.2 + 136.1 + 4.0 + 2.6] \times 2$
= $219.0 \times 2 = 438 \text{ W}$
Ventilation: $1.20 \times 1006 \times 0.080 \times 2 = 193 \text{ W}$
 $Q_{\text{heat_total}} = 438 + 193 = 631 \text{ W} \approx 0.63 \text{ kW}$

Heating load = 0.63 kW — NEGLIGIBLE. Cooling-only DX system fully justified.

6. REFRIGERATION SYSTEM DESIGN

6.1 System Selection

System type: Variable Refrigerant Flow (VRF) Multi-Split Direct Expansion (DX) system with one outdoor condensing unit and four indoor fan-coil units (FCUs). This configuration is standard practice for marine vessels in this tonnage range.

Refrigerant: R-410A (HFC, ASHRAE Group A1, ODP = 0, GWP = 2,088). R-32 is considered as a lower-GWP alternative (GWP = 675) for future re-specification.

6.2 Operating Conditions — Cycle Parameters

State Point	Condition	Temperature (°C)	Pressure (kPa)	Enthalpy (kJ/kg)
State 1 — Compressor Inlet	Superheated vapour	12 (7 + 5 SH)	1,084	447
State 2 — Compressor Outlet	Hot discharge gas	≈ 86 (actual)	2,760	504
State 3 — Condenser Outlet	Subcooled liquid	45 (50 – 5 SC)	2,760	271
State 4 — Evaporator Inlet	Wet mixture (flash)	7 (T _{evap})	1,084	271

R-410A property data sourced from ASHRAE Refrigerant Tables (2017 Fundamentals). Evaporating temperature 7 °C is set deliberately low (cf. typical 10 °C) to handle the high latent load (SHR = 0.545).

6.3 P-h Diagram States & Refrigerating Effect

$$\text{Refrigerating Effect (RE)} = h_1 - h_4$$

$$\text{RE} = 447 - 271 = 176 \text{ kJ/kg}$$

6.4 Refrigerant Mass Flow Rate

$$\dot{m} = Q_{\text{evap}} / \text{RE}$$

$$\dot{m} = 13,562 \text{ W} / 176,000 \text{ J/kg}$$

$$\dot{m} = 0.07705 \text{ kg/s}$$

$$\text{Refrigerant mass flow rate } \dot{m} = 0.0771 \text{ kg/s}$$

6.5 Compressor Sizing

Isentropic efficiency of a modern scroll compressor at this duty: $\eta_{\text{isen}} = 0.75$ (typical, ASHRAE rated conditions).

$$\text{Isentropic compressor work: } w_{\text{isen}} = h_{2s} - h_1$$

$$h_{2s} \text{ (isentropic discharge at } 50^\circ\text{C condensing)} = 490 \text{ kJ/kg}$$

$$w_{\text{isen}} = 490 - 447 = 43 \text{ kJ/kg}$$

$$\text{Actual enthalpy at discharge: } h_2 = h_1 + w_{\text{isen}} / \eta_{\text{isen}}$$

$$h_2 = 447 + 43/0.75 = 447 + 57.3 = 504.3 \text{ kJ/kg}$$

$$\text{Actual compressor power: } W_{\text{comp}} = \dot{m} \times (h_2 - h_1)$$

$$W_{\text{comp}} = 0.0771 \times (504.3 - 447) = 0.0771 \times 57.3$$

$$W_{\text{comp}} = 4,418 \text{ W} \approx 4.42 \text{ kW}$$

6.6 Coefficient of Performance (COP)

$$\text{COP}_{\text{cooling}} = Q_{\text{evap}} / W_{\text{comp}}$$

$$\text{COP} = 13,562 / 4,418$$

COP = 3.07 (air-cooled, tropical conditions — acceptable for marine application)

6.7 Condenser Duty

$$Q_{\text{cond}} = Q_{\text{evap}} + W_{\text{comp}}$$

$$Q_{\text{cond}} = 13,562 + 4,418$$

$Q_{\text{cond}} = 17,980 \text{ W} \approx 18.0 \text{ kW}$

6.8 Compression Ratio

$$CR = P_{\text{cond}} / P_{\text{evap}} = 2,760 / 1,084$$

Compression Ratio CR = 2.55 (within recommended range 2–4 for scroll compressor)

6.9 Condenser Air Flow

The condensing unit is roof-mounted (aft deck) with axial propeller fans. Condenser air temperature rise across the coil is assumed at $\Delta T_{\text{air}} = 10^\circ\text{C}$.

$$\dot{m}_{\text{air}} = Q_{\text{cond}} / (c_p \times \Delta T_{\text{cond}})$$

$$\dot{m}_{\text{air}} = 17,980 / (1,006 \times 10) = 17,980 / 10,060$$

$$\dot{m}_{\text{air}} = 1.787 \text{ kg/s}$$

$$V_{\text{air}} = \dot{m}_{\text{air}} / \rho = 1.787 / 1.20$$

Condenser air flow = 1.49 m³/s = 5,364 m³/h → 2 × fans, 2,682 m³/h each

6.10 Refrigeration Cycle Summary

Parameter	Symbol	Value	Unit
Design cooling load	Q_{evap}	13.6	kW
Refrigerating effect	RE	176	kJ/kg
Refrigerant mass flow	$\dot{m}$	0.0771	kg/s
Compressor power (actual)	W_{comp}	4.42	kW
Condenser duty	Q_{cond}	18.0	kW
COP (cooling)	COP	3.07	—
Compression ratio	CR	2.55	—
Evaporating temperature	T_{evap}	7	°C
Condensing temperature	T_{cond}	50	°C
Refrigerant	—	R-410A	—

7. FAN COIL UNIT SIZING

7.1 Zone Load Distribution

Zone loads are distributed from the total using floor area weighting, with manual adjustments for zones with disproportionate internal loads (saloon has more equipment; galley has cooking load).

7.2 Supply Air Temperature & Flow Rates

Supply air temperature is set at 12 °C. At SHR = 0.545, this requires a coil apparatus dew point (ADP) of approximately 8 °C, achievable with a 7 °C evaporating temperature.

$$\dot{m}_{\text{air_zone}} = Q_{\text{sens_zone}} / (c_p \times (T_{\text{room}} - T_{\text{supply}}))$$
$$V_{\text{zone}} = \dot{m}_{\text{air_zone}} / \rho_{\text{air}} = \dot{m} / 1.20 \rightarrow \text{L/s}$$

7.3 Zone-by-Zone FCU Calculation

Zone 1: Main Saloon (19.4 m², 5 persons)

Envelope share (35% of envelope total):	$0.35 \times (1122+1196+816+47+245) = 0.35 \times 3,426 = 1,199 \text{ W}$
Solar gain (saloon-specific):	180 W
Occupancy sensible (5 × 75W):	375 W
Lighting (10 × 19.4):	194 W
Equipment (AV, TV):	200 W
Ventilation share (37.5%):	$0.375 \times 5,232 = 1,962 \text{ W}$
Infiltration share (35%):	$0.35 \times 1,071 = 375 \text{ W}$
Zone sensible sub-total:	4,485 W
Zone latent (35% of total latent):	$0.35 \times 6,169 = 2,159 \text{ W}$
Zone total:	6,644 W
After 10% safety factor: safety already in total	$6,644 \times 1.10 \div 1 = \text{note:}$
Zone fraction of design load:	$6,644/12,329 \times 13,562 =$
5,307 W (adjusted)	
Supply air flow: $Q_s / (c_p \times \Delta T) = 4,485/(1006 \times 12) = 0.372 \text{ kg/s} = 0.310 \text{ m}^3/\text{s} = 310 \text{ L/s}$	

Zone 1 (Saloon): Load ≈ 5.3 kW → Select 5.5 kW ceiling cassette FCU | Airflow: 310 L/s

Zone 2: Master Cabin (15.1 m², 2 persons)

Envelope share (27%):	$0.27 \times 3,426 = 925 \text{ W}$
Solar (porthole, proportioned):	32 W
Occupancy sensible (2 × 75W):	150 W
Lighting (10 × 15.1):	151 W
Equipment (TV, charging):	100 W
Ventilation share (25%):	$0.25 \times 5,232 = 1,308 \text{ W}$
Infiltration share (27%):	$0.27 \times 1,071 = 289 \text{ W}$
Zone sensible sub-total:	2,955 W
Zone total (sensible + latent):	$2,955 + 0.27 \times 6,169 =$
2,955 + 1,666 = 4,621 W	
Zone fraction of design load:	$4,621/12,329 \times 13,562 =$
5,084 → conservative	
Use zone sensible for airflow:	$2,955/(1006 \times 12) = 0.245$
kg/s = 0.204 m³/s = 204 L/s	

Zone 2 (Master): Load ≈ 3.5 kW → Select 3.5 kW high-wall FCU | Airflow: 204 L/s

Zone 3: Cabin 2 (12.9 m², 2 persons)

Using same methodology, area fraction = $12.9/56.0 = 23\%$

Zone sensible: 2,530 W Zone total: 3,950 W

Air flow: $2,530/(1006 \times 12) = 0.210 \text{ m}^3/\text{s} = 175 \text{ L/s}$

Zone 3 (Cabin 2): Load $\approx 3.0 \text{ kW}$ → Select 3.5 kW high-wall FCU | Airflow: 175 L/s

Zone 4: Galley (8.6 m², 1 crew, cooking)

Zone sensible includes cooking load: 300 W extra

Envelope share (15.4%) + cooking: $527 + 150 + 86 + 100 + 300 + 805 + 165 = 2,133 \text{ W}$

Zone total (sensible + latent): $2,133 + 0.154 \times 6,169 = 2,133 + 950 = 3,083 \text{ W}$

Air flow: $2,133/(1006 \times 12) = 0.177 \text{ m}^3/\text{s} = 147 \text{ L/s}$

Zone 4 (Galley): Load $\approx 3.1 \text{ kW}$ → Select 3.5 kW concealed FCU | Airflow: 147 L/s

7.4 FCU Selection Summary

Zone	Zone Load (kW)	Airflow (L/s)	T _{supply} (°C)	Selected FCU (kW)	FCU Type
Main Saloon	5.3	310	12	5.5	Ceiling cassette
Master Cabin	3.5	204	12	3.5	High-wall split
Cabin 2	3.0	175	12	3.5	High-wall split
Galley	3.1	147	12	3.5	Concealed duct
Total installed	14.9	836	—	16.0	—

Installed capacity (16.0 kW) exceeds design load (13.6 kW) by 17.6%. This margin accounts for performance degradation under sustained tropical operation, coil fouling, and future load growth. The VRF controller will part-load the system under normal conditions, maintaining efficiency.

8. DISTRIBUTION SYSTEM DESIGN

8.1 Ductwork — Main Saloon Mini-Duct FCU

Zones 2, 3, and 4 use cassette/wall FCUs with no separate ductwork. Zone 1 (Saloon) uses a mini-duct concealed unit feeding three ceiling diffusers through short GRP-lined galvanised steel ducts.

8.1.1 Design Parameters

Parameter	Value	Standard / Basis
Target supply duct velocity	3.0 m/s max	Marine quiet — ASHRAE Ch. 21
Target branch duct velocity	2.5 m/s max	Low-noise residential
Friction method	Equal friction	ASHRAE duct design
Duct material	Galvanised steel, insulated	12 mm foam-faced liner
Total saloon air flow	310 L/s = 0.310 m³/s	From Zone 1 calculation

8.1.2 Main Supply Duct (Full Flow, 310 L/s)

$$\text{Required area} = Q / v = 0.310 / 3.0 = 0.1033 \text{ m}^2$$

$$\text{Rectangular duct: } 400 \text{ mm} \times 260 \text{ mm} = 0.1040 \text{ m}^2 \quad \checkmark$$

$$\text{Hydraulic diameter: } D_h = 4A/P = 4 \times 0.1040 / (2 \times 0.660) = 0.316 \text{ m}$$

$$\text{Actual velocity} = 0.310 / 0.1040 = 2.98 \text{ m/s} \quad \checkmark (\leq 3.0 \text{ m/s})$$

8.1.3 Branch Ducts (3 off, equal flow = 103 L/s each)

$$\text{Required area} = 0.1033 / 3 = 0.0344 \text{ m}^2 \quad (\text{at } 3.0 \text{ m/s})$$

$$\text{At } 2.5 \text{ m/s: Area} = 0.1033 / (3 \times 2.5) = 0.01377 \text{ m}^2$$

$$\text{Rectangular: } 150 \text{ mm} \times 95 \text{ mm} = 0.01425 \text{ m}^2 \rightarrow \text{use } 150 \times 100 \text{ mm}$$

$$\text{Actual velocity} = 0.1033 / (3 \times 0.0150) = 2.30 \text{ m/s} \quad \checkmark$$

8.1.4 Pressure Loss — Darcy-Weisbach Method

Friction factor $f = 0.020$ (smooth galvanised steel, ASHRAE Moody chart, $Re > 10,000$).

$$\Delta P_f = f \times (L/D_h) \times (\rho \times v^2 / 2)$$

$$\text{Main duct (L = 5.0 m, } D_h = 0.316 \text{ m, } v = 2.98 \text{ m/s):}$$

$$\Delta P_f = 0.020 \times (5.0 / 0.316) \times (1.20 \times 2.98^2 / 2)$$

$$= 0.020 \times 15.82 \times 5.33 = 1.69 \text{ Pa}$$

$$\text{Fittings (2 elbows } K=0.50, 1 \text{ tee } K=1.00):$$

$$\Delta P_{fit} = (2 \times 0.50 + 1.00) \times 5.33 = 2.0 \times 5.33 = 10.66 \text{ Pa}$$

$$\text{Main duct total: } 1.69 + 10.66 = 12.35 \text{ Pa}$$

$$\text{Branch duct (L = 2.5 m, } D_h = 0.12 \text{ m, } v = 2.30 \text{ m/s):}$$

$$\Delta P_f = 0.020 \times (2.5 / 0.12) \times (1.20 \times 2.30^2 / 2)$$

$$= 0.020 \times 20.83 \times 3.174 = 1.32 \text{ Pa}$$

$$\text{Elbow: } 0.50 \times 3.174 = 1.59 \text{ Pa}$$

$$\text{Branch total: } 1.32 + 1.59 = 2.91 \text{ Pa}$$

$$\text{Diffuser pressure drop: } 12.0 \text{ Pa} \quad (\text{manufacturer, marine low-profile})$$

$$\text{Return air path: } 12.0 \text{ Pa}$$

$$\text{Coil + foam filter: } 80.0 \text{ Pa} \quad (\text{manufacturer rated data})$$

$$\text{Total ESP} = 12.35 + 2.91 + 12.0 + 12.0 + 80.0 = 119.3 \text{ Pa}$$

Specify FCU Zone 1 with External Static Pressure (ESP) ≥ 120 Pa

8.2 Refrigerant Piping — Main Lines

The condensing unit is located on the aft roof deck. Refrigerant lines run below the deck through insulated conduit to each FCU. Estimated main run length: 12 m. Pipe material: annealed copper, Table ACR (ASTM B280).

8.2.1 R-410A Property Data at Operating Conditions

State	T (°C)	Phase	Density (kg/m³)	Spec. Volume (m³/kg)
Suction line (superheated)	12	Vapour	46.2	0.02165
Discharge line	86	Superheated gas	—	use manufacturer
Liquid line (subcooled)	45	Liquid	910	0.001099

8.2.2 Suction Line (Main — all FCUs combined)

ASHRAE recommends suction line velocity of 4–10 m/s (horizontal) to ensure oil return to compressor.

$\text{Volume flow} = \dot{m} \times v_g = 0.0771 \times 0.02165 = 0.001669 \text{ m}^3/\text{s}$
 $\text{Target velocity } v = 7 \text{ m/s} \rightarrow A = Q/v = 0.001669/7 = 0.000239 \text{ m}^2$
 $D = \sqrt[4]{(4A/\pi)} = \sqrt[4]{(4 \times 0.000239/\pi)} = 0.01744 \text{ m} = 17.4 \text{ mm}$
 Select: 7/8" OD (22.2 mm) copper ACR tube $\rightarrow$ ID $\approx$ 20.0 mm
 $\text{Actual area} = \pi \times 0.010^2 = 0.000314 \text{ m}^2$
 $\text{Actual velocity} = 0.001669 / 0.000314 = 5.32 \text{ m/s} \quad \checkmark \quad (\text{within } 4\text{--}10 \text{ m/s})$

Suction line: 7/8" OD (22.2 mm) copper — actual velocity 5.32 m/s $\checkmark$

8.2.3 Liquid Line (Main)

Liquid line velocity should be 0.5–1.5 m/s to avoid flashing and ensure subcooled liquid reaches the expansion device.

$\text{Volume flow} = \dot{m} / \rho_{\text{liq}} = 0.0771 / 910 = 0.0000848 \text{ m}^3/\text{s}$
 $\text{Target velocity } v = 1.0 \text{ m/s} \rightarrow A = 0.0000848 \text{ m}^2$
 $D = \sqrt[4]{(4 \times 0.0000848/\pi)} = 0.01039 \text{ m} = 10.4 \text{ mm}$
 Select: 3/8" OD (9.52 mm) copper ACR tube $\rightarrow$ ID $\approx$ 8.0 mm
 $\text{Actual area} = \pi \times 0.004^2 = 0.0000503 \text{ m}^2$
 $\text{Actual velocity} = 0.0000848 / 0.0000503 = 1.69 \text{ m/s} \quad \checkmark \quad (\text{acceptable for liquid line})$

Liquid line: 3/8" OD (9.52 mm) copper — actual velocity 1.69 m/s $\checkmark$

8.2.4 Pressure Drop Verification — Suction Line

Allowable pressure drop in suction line is conventionally limited to an equivalent saturation temperature drop of 1 °C to preserve system efficiency.

$\Delta P_f = f \times (L/D) \times (\rho_v \times v^2/2)$
 $f = 0.020 \quad (\text{smooth copper, turbulent})$
 $L = 12 \text{ m} \quad (\text{main run, worst-case distance to Zone 1 FCU})$
 $D = 0.020 \text{ m} \quad (\text{inner diameter})$
 $\rho_v = 46.2 \text{ kg/m}^3 \quad v = 5.32 \text{ m/s}$
 $\Delta P = 0.020 \times (12/0.020) \times (46.2 \times 5.32^2/2)$
 $= 0.020 \times 600 \times 653.4 = 7,841 \text{ Pa} = 7.84 \text{ kPa}$
 $\text{Equivalent } \Delta T = \Delta P \times (dT/dP) |_{\text{sat}}$
 For R-410A at 7°C: $dT/dP \approx 0.036 \text{ }^\circ\text{C/kPa}$
 $\Delta T_{\text{equiv}} = 7.84 \times 0.036 = 0.28 \text{ }^\circ\text{C} < 1.0 \text{ }^\circ\text{C allowable} \quad \checkmark$

Suction line pressure drop: 7.84 kPa $\rightarrow$ equivalent $\Delta T = 0.28 \text{ }^\circ\text{C} \quad \checkmark \quad \text{ACCEPTABLE}$

8.3 Piping Summary

Line	Service	Size (OD)	ID	Velocity (m/s)	Material
Suction main	All FCUs to CU	7/8" (22.2 mm)	20.0 mm	5.32	Copper ACR
Suction branch (Zones 2-4)	Each FCU	1/2" (12.7 mm)	10.9 mm	4–6	Copper ACR
Liquid main	CU to all FCUs	3/8" (9.52 mm)	8.0 mm	1.69	Copper ACR
Liquid branch	Each FCU	1/4" (6.35 mm)	4.8 mm	<1.5	Copper ACR
Hot gas (discharge)	CU only	5/8" (15.9 mm)	14.0 mm	per OEM	Copper ACR

9. ELECTRICAL LOAD SCHEDULE & SYSTEM SUMMARY

9.1 HVAC Electrical Load Schedule

Equipment	Rating	Load (kW)	Remarks
Condensing unit — scroll compressor	1 × 5.5 kW motor	4.42	Running load at duty cycle
Condenser fans	2 × 250 W axial	0.50	Continuous run
FCU 1 — Saloon (5.5 kW unit)	Fan motor	0.18	High speed
FCU 2 — Master Cabin (3.5 kW)	Fan motor	0.12	High speed
FCU 3 — Cabin 2 (3.5 kW)	Fan motor	0.10	High speed
FCU 4 — Galley (3.5 kW)	Fan motor	0.10	High speed
Controls / BMS / room stats	DC powered	0.05	All zones
Total HVAC operating load	—	5.47 kW	All running simultaneously

9.2 Generator Adequacy Check

Available power = Generator kVA × power factor = 15 × 0.80 = 12.0 kW

HVAC system demand: 5.47 kW (46% of available)

HVAC compressor start load: $5.47 + 4.42 \times (\text{LRA factor } 2.5 - 1) = 5.47 + 6.63 = 12.1 \text{ kW}$

Note: soft-starter or inverter-driven compressor is specified to limit starting current to 120% of full-load current, bringing start demand to approximately $5.47 + 4.42 \times 0.20 = 6.35 \text{ kW}$.

Remaining load budget (non-HVAC): $12.0 - 5.47 = 6.53 \text{ kW}$ ✓

Generator adequacy CONFIRMED — 15 kVA is sufficient with inverter-driven compressor

9.3 System Summary Table

Design Parameter	Value	Unit
Vessel	MV Anamika, 45 ft houseboat	
Location	Alappuzha, Kerala (9.5°N)	
Total conditioned area	56.0	m ²
Total design cooling load	13.6	kW
Cooling load in TR	3.87	TR
Design sensible heat ratio	0.545	—
Refrigerant	R-410A	
Evaporating temperature	7	°C
Condensing temperature	50	°C
Coefficient of Performance	3.07	—
Compressor power	4.42	kW
Total installed FCU capacity	16.0	kW
Saloon FCU airflow	310	L/s
Total fresh air supply	80	L/s

Suction line size (main)	7/8" OD copper	
Liquid line size (main)	3/8" OD copper	
Total HVAC electrical load	5.47	kW
Generator capacity	15	kVA
Design standard (ventilation)	ASHRAE 62.1	
Design standard (load calc)	ASHRAE Fundamentals / CIBSE	

9.4 Engineering Notes

1. CORROSION PROTECTION: All refrigerant copper pipework to be cleaned, pressure-tested to 40 bar (R-410A design), and wrapped with closed-cell PE foam insulation (19 mm wall) to prevent condensation and corrosion in the marine environment.
2. VIBRATION ISOLATION: Condensing unit mounted on anti-vibration mounts (natural frequency < 3 Hz) to prevent structure-borne noise transmission to sleeping cabins.
3. CONDENSATE DRAINAGE: All FCU drip trays to be GRP-lined and drain via 25 mm PVC pipes with P-traps to hull drain sea-cocks. Adequate condensate volume: approximately 8–10 litres/hour at design conditions.
4. REFRIGERANT LEAK DETECTION: R-410A leak detector sensor (Danfoss-type) to be installed in the engine room where condensing unit lines enter the hull, with alarm output to wheelhouse panel.
5. FUTURE RE-SPECIFICATION: R-32 (GWP = 675 vs 2,088 for R-410A) is recommended for the next vessel build or refit to reduce environmental impact. All pipe sizing calculated above remains applicable as the volumetric refrigerating capacity of R-32 is higher, resulting in smaller pipe diameters.